

Requirements: data-driven camera calibration for light field camera systems

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Requirements for the Cross-Shaped Camera Array Calibration

Based on the described use case and the methods presented in the reference paper (LiFCal), the following requirements define the functional scope and measurable performance goals for the implementation in this Bachelor's thesis. The system assumes a predominantly static scene during calibration and monitoring, while allowing the camera array to take different poses across captured image sets.

Functional Requirements

- **FR1 – Initial Reference Calibration (Target-based):** The system shall perform a one-time, checkerboard-based initial calibration using OpenCV to estimate intrinsic and extrinsic parameters for the cross-shaped camera array. The resulting calibration shall be stored as the reference (baseline) calibration.
- **FR2 – Periodic Health Monitoring (Parameter-based):** The system shall periodically (e.g. every 3 minutes) capture a small image set during normal operation and compute calibration health metrics based on: (i) intrinsic deviation from the baseline, (ii) extrinsic deviation from the baseline, and (iii) normalized parameter drift between consecutive monitoring windows.
- **FR3 – Health Score Computation:** The system shall combine the normalized intrinsic/extrinsic deviations and drift metrics into a single scalar health score representing the current calibration quality (Score: [0 - 100]).
- **FR4 – Automatic Trigger for Target-Free Recalibration:** If the health score exceeds a predefined threshold for a configurable number of consecutive monitoring windows, the system shall automatically initiate Phase 3 (target-free online recalibration).
- **FR5 – Target-Free Online Recalibration (LiFCal-based):** The system shall collect a larger image set from recent normal operation and perform a LiFCal-based parameter estimation on this extended image set without using calibration targets (no checkerboard).
- **FR6 – Acceptance and Rollback Rule:** The updated calibration shall be adopted only if the post-recalibration health score improves compared to the previously active calibration; otherwise, the system shall revert to the last known-good calibration.
- **FR7 – Logging and Traceability:** The system shall log all health-check runs and recalibration runs, including timestamps, used image sets, calibration parameters, health scores, and acceptance decisions, to support evaluation and reproducibility.

Non-Functional Requirements

- **NFR1 – Portability:** The calibration framework shall be implemented in Python/C++ with OpenCV and remain adaptable to different camera configurations (e.g., Raytrix R5, R25, or similar).

- **NFR2 – Modularity:** Each component (data acquisition, health monitoring, bundle adjustment, logging) shall be designed as a separate, reusable module.
- **NFR3 – Reproducibility:** Calibration and monitoring results shall be exportable in a standardized format (e.g., JSON) to ensure consistent evaluation.
- **NFR4 – Robustness to Scene Variation:** The calibration process shall tolerate moderate lighting and texture variations without significant loss of accuracy (health score drop ≤ 10 points).